

PART 1

1. **[3 points]** Consider the execution of a MapReduce algorithm on a cluster with 10 machines, each equipped with a RAM of 8 GB and a disk of 128 GB. Before and after the Map and Reduce Phases of each round, data are stored into a HDFS built on the union of the disks. Let M_L and M_A be the algorithm's local and aggregate space. What are the maximum values (in GB) for M_L and M_A which ensure a successful execution of the algorithm?
2. **[3 points]** The coreset-based MapReduce algorithm for k-means computes the final centers of the clusters by running a sequential algorithm $\mathcal{A}$ on a weighted coreset T .
 - (a) How are the points in T computed?
 - (b) For each point $q \in T$, what does its weight $w(q)$ represent?
3. **[4 points]** Consider the execution of Reservoir Sampling on an unbounded stream $\Sigma = x_1, x_2, \dots$. For $t > m$, let S_t be the set of m elements maintained by the algorithm after processing x_t , and suppose that x_t has been added to S_t . For an element x_i with $i < t$, determine probability that $x_i \in S_t$, given that $x_t \in S_t$.
4. **[3 points]** Let P be a set of points from a metric space (M, d) . Define the (c, r) -Approximate Near Neighbor Search problem and explain how it differs from the exact r -Near Neighbor Search problem.

PART 2

1. **[7 points]** Let P be a dataset containing N outcomes of slot machines, represented by the key-value pairs $(i, (s_i, o_i))$, for $0 \leq i < N$, where s_i is a slot machine, and o_i is an outcome of the machine. An outcome o is called *frequent for a slot machine s* if P contains $\geq N/50$ pairs with value (s, o) , and a slot machine s is called *biased* if there is at least a frequent outcome o for s .
 - (a) Design a 3-round MapReduce algorithm that for every biased slot machine s returns exactly one pair (s, null) . (**HINT:** in the first 2 rounds, compute the pairs (s, o) where o is frequent for s .)
 - (b) Analyze the local and aggregate space required by your algorithm.

For full score, your algorithm must require $o(N)$ local space and $O(N)$ aggregate space.

2. **[6 points]** Consider a stream $\Sigma = x_1, x_2, \dots, x_n$ of n measurements from sensors. Each measurement x_i is a pair (k_i, w_i) , where k_i is the ID of a sensor and w_i is the value of the measurement (an integer). For a given sensor u occurring in Σ define

$$f_u = \sum_{(k_i, w_i) \in \Sigma : k_i = u} w_i,$$

i.e., the aggregate measurements taken by u . Briefly devise a space-efficient estimator for $\sum_u (f_u)^2$, where the sum is over all sensors occurring in the stream. Give an intuition about whether the estimator is biased or unbiased.

Total time: 2 hours